SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER APPLICATIONS

TIRUCHIRAPPALLI

RECORD WORK

PCA25C04J – DATA ANALYSIS UNSING PYTHON PROGRAMMING

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Name of the Student :

Semester / Year : **II / I**

Programme : **M.C.A**

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER APPLICATIONS

REGISTER NO._____

BONAFIDE CERTIFICATE

Certified as the bonafide record of work done by RA25322410500 Register Number 2 of 1 (Semester/Year), MASTER OF COMPUTER APPLICATIONS **Programme** in the practical course **PCA25C04J – DATA ANALYSIS USING PYTHON PROGRAMMING** at **SRM Institute of Science and Technology, Tiruchirappalli Campus** during the Academic Year, 2025-2026.

Faculty In-charge **Head of the Department**

Submitted for the End Semester Examination held on _____

Examiner-1 **Examiner-2**

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Program No: 1 A File Name: Ex. No: Date: _____	IMPLEMENTATION OF RECURSION IN PYTHON (FACTORIAL)

AIM:

Write a Python program to find factorial calculation using recursion.

ALGORITHM:

Step 1: Get a positive integer input (n) from the user.

Step 2: 0! = 1 & 1! = 1

Step 3: n! = n x (n-1)!

Step 4: The factorial of a negative integer is undefined (i.e., (-n)! is not defined).

Step 5: The factorial function grows very quickly, and for large values of n, it can overflow the capacity of standard data types like integers.

Step 6: Iterate from 1 to n using a for loop (for loop is used to increment the number up to the given input)

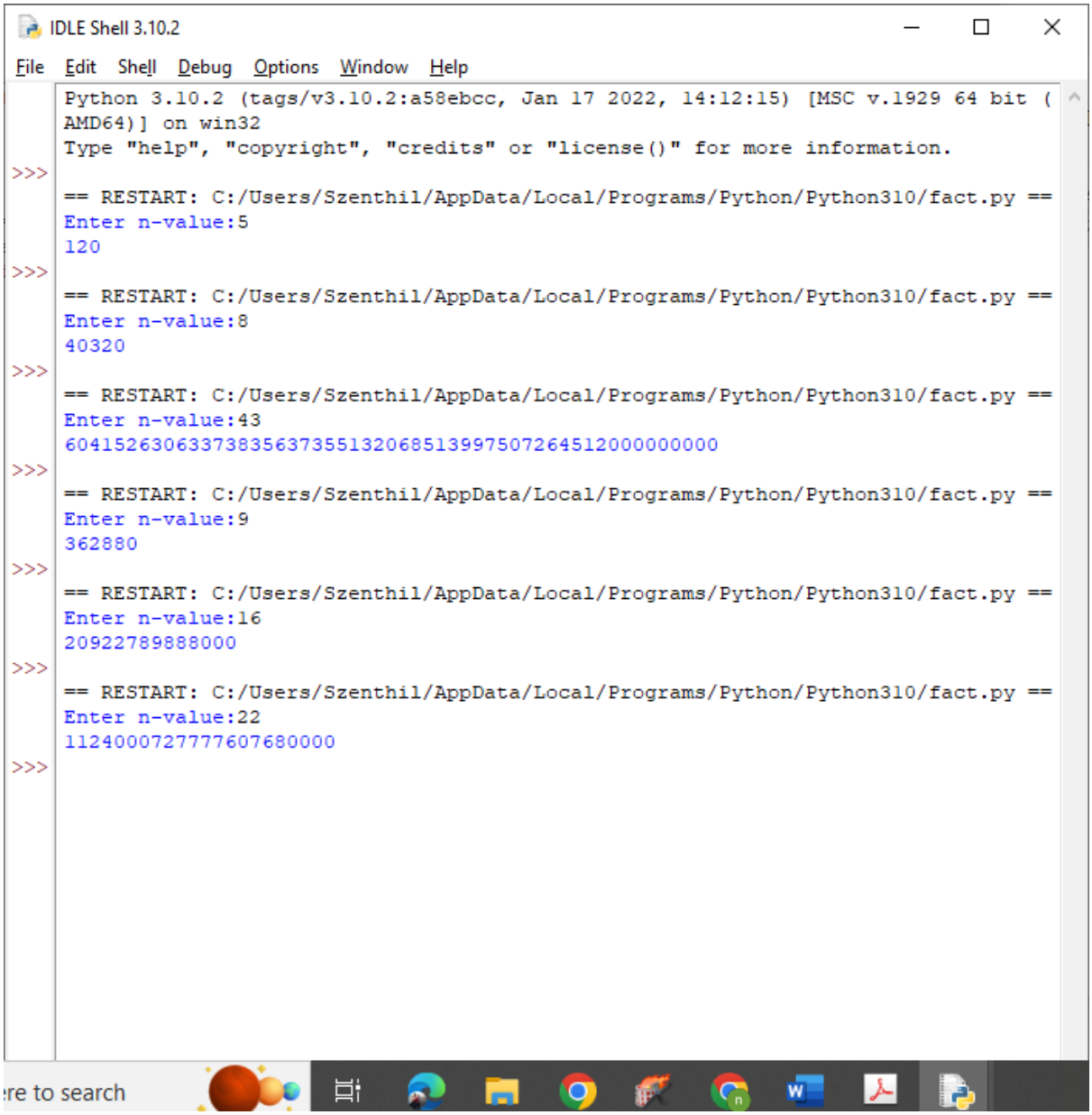
Step 7: Using the below formula, calculate the factorial of a number = f*i.

Step 8: Print the output i.e., the calculated factorial.

PROGRAM:

```
def factorial(n):
    if (n==0 or n==1):
        return 1
    else:
        return n * factorial(n-1)
n = int(input('Enter n-value:'))
res = factorial(n)
print(res)
```

OUTPUT:



```
IDLE Shell 3.10.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/fact.py ==
Enter n-value:5
120
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/fact.py ==
Enter n-value:8
40320
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/fact.py ==
Enter n-value:43
604152630633738356373551320685139975072645120000000000
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/fact.py ==
Enter n-value:9
362880
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/fact.py ==
Enter n-value:16
20922789888000
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/fact.py ==
Enter n-value:22
1124000727777607680000
>>>
```

RESULT:

Thus, the python program for computing factorial is successfully completed.

Program No: 1 B File Name: Ex. No: Date: _____	PYTHON PROGRAM TO CHECK PRIME NUMBER

AIM:

To write a Python Program for to check whether the given number is Prime number or not.

ALGORITHM:

Step 1: Initialize a for loop starting from 2 and ending at the number

Step 2: Check if the number is divisible by 2

Step 3: Repeat till the number -1 is checked for

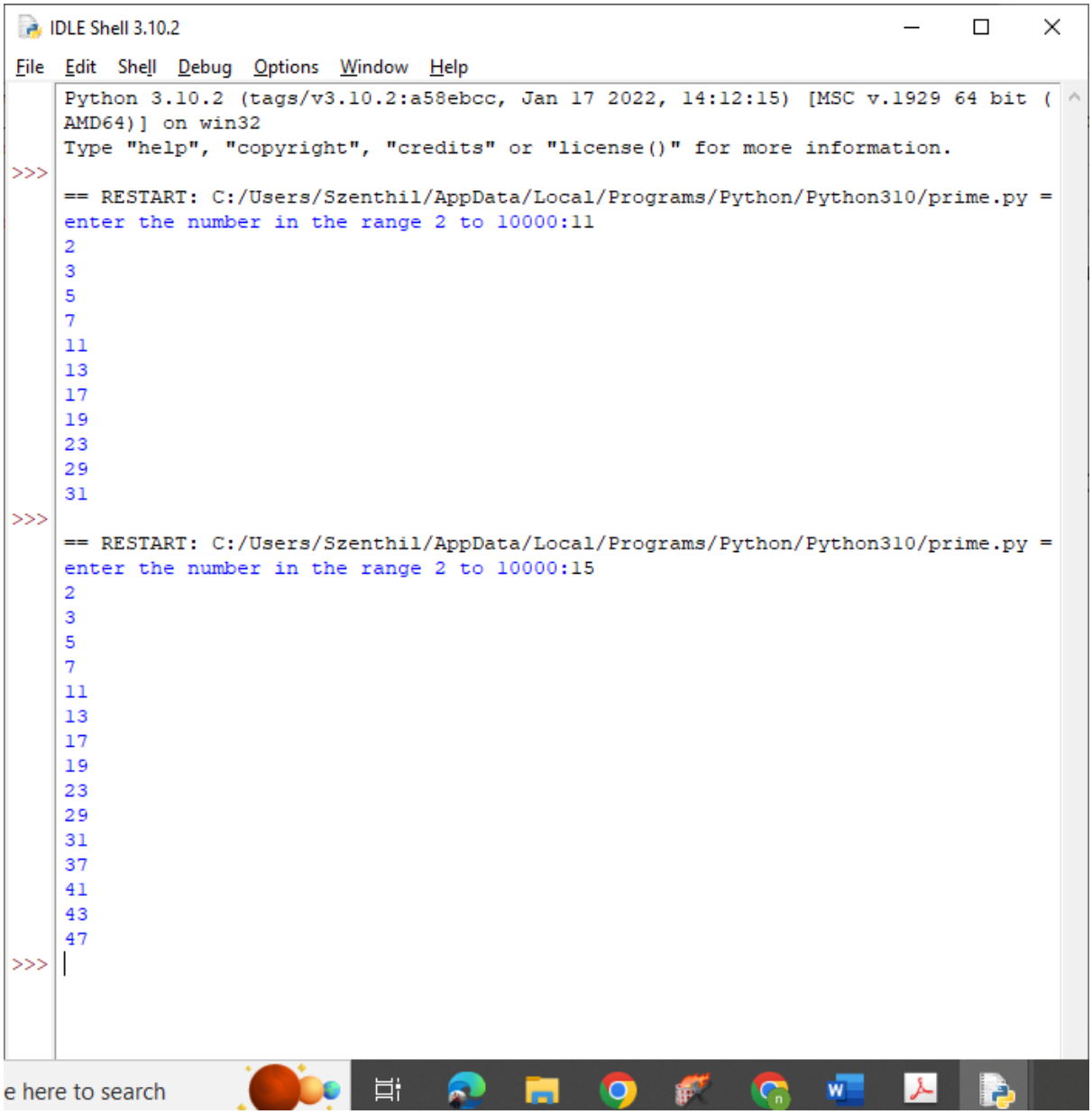
Step 4: In case, the number is divisible by any of the numbers, the number is not prime

Step 5: Else, it is a prime number

PROGRAM:

```
number=int(input("enter the number in the range 2 to 10000:"))
pNumbers=[]
for num in range(2,10000):
    if all(num%i!=0 for i in range(2,num)):
        pNumbers.append(num)
for i in range(number):
    print(pNumbers[i])
```

OUTPUT:



RESULT:

Thus, the python program for finding the prime number is successfully completed.

Program No: 1 C File Name: Ex. No: Date: _____	ARITHMETIC OPERATORS IN PYTHON

AIM:

Write a Python program using Arithmetic Operators to perform mathematical operations like addition, subtraction, multiplication, division, etc.

ALGORITHM:

Step 1: Get two integers as inputs: a & b.

Step 2: Give the input value for a.

Step 3: Give the input value for b.

Step 4: Print the (Sum, Subtraction, Multiplication, Division, Floor Division, Modulo and Power)

PROGRAM:

```
a = int(input("Enter the First Number: "))

b = int(input("Enter the Second Number: "))

print ('Sum: ', a + b)                # addition

print ('Subtraction: ', a - b)         # subtraction

print ('Multiplication: ', a * b)      # multiplication

print ('Division: ', a / b)            # division

print ('Floor Division: ', a // b)     # floor division

print ('Modulo: ', a % b)              # modulo

print ('Power: ', a ** b)              # a to the power b
```

OUTPUT:

```
IDLE Shell 3.10.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (AM
D64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/operators.py
Enter the First Number: 15
Enter the Second Number: 23
Sum:  38
Subtraction:  -8
Multiplication:  345
Division:  0.6521739130434783
Floor Division:  0
Modulo:  15
Power:  11222741464018821716308593 75
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/operators.py
Enter the First Number: 67
Enter the Second Number: 79
Sum:  146
Subtraction:  -12
Multiplication:  5293
Division:  0.8481012658227848
Floor Division:  0
Modulo:  67
Power:  1819321338736902125229335845622505507704549101261646195761894037378497941 4
4547153562345267813354228846795668175133962580161207340795515193091 4603
>>>
Ln: 25  Col: 0
```

RESULT:

Thus, the python program using arithmetic operators is successfully completed.

Program No: 1 D File Name: Ex. No: Date: _____	PYTHON PROGRAM FOR DIVISIBLE NUMBERS BY 2 AND 3

AIM:

To write a Python Program to perform Divisible numbers by 2 and 3 of a given number.

ALGORITHM:

Step1: Start

Step2: Read the value for num.

Step3: Check if the last digit of the number is even.

Step4: If the number is even, then the number is divisible by 2.

Step5: Check if the sum of the digits of the number is divisible by 3.

Step6: If it is, then the number is divisible by 3.

Step7: Print the value of num.

Step8: Stop

PROGRAM:

```
num=8
print ("num = ",num)
if num%2==0:
  if num%3==0:
    print ("Divisible by 3 and 2")
  else:
    print ("divisible by 2 not divisible by 3")
else:
  if num%3==0:
    print ("divisible by 3 not divisible by 2")
  else:
    print ("not Divisible by 2 not divisible by 3")
```

OUTPUT:

```
IDLE Shell 3.10.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========================================= RESTART: C:/Users/Szenthil/AppData/Local
/Programs/Python/Python310/div by 2 and 3.py ====================================
===
num =  8
divisible by 2 not divisible by 3
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/div by 2 an
d 3.py
num =  15
divisible by 3 not divisible by 2
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/div by 2 an
d 3.py
num =  5
not Divisible by 2 not divisible by 3
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/div by 2 an
d 3.py
num =  12
Divisible by 3 and 2
>>>
                                                                    Ln: 14  Col: 37
```

RESULT:

Thus, the python program for perform Divisible numbers by 2 and 3 of a given number is successfully completed.

Program No: 1 E File Name: Ex. No: Date: _____	PYTHON PROGRAM TO COMPUTE THE GCD OF TWO NUMBERS

AIM:

To write a Python Program to compute the GCD of two numbers.

ALGORITHM:

Step1: Start

Step2: Read the value of first number a.

Step3: Read the Value of Second number b.

Step4: Calculate rem=a%b.

Step5: Repeat the steps 6 to 7 until rem!=0.

Step6: Assign a=b.

Step7: Assign b=rem.

Step8: Calculate rem=a%b.

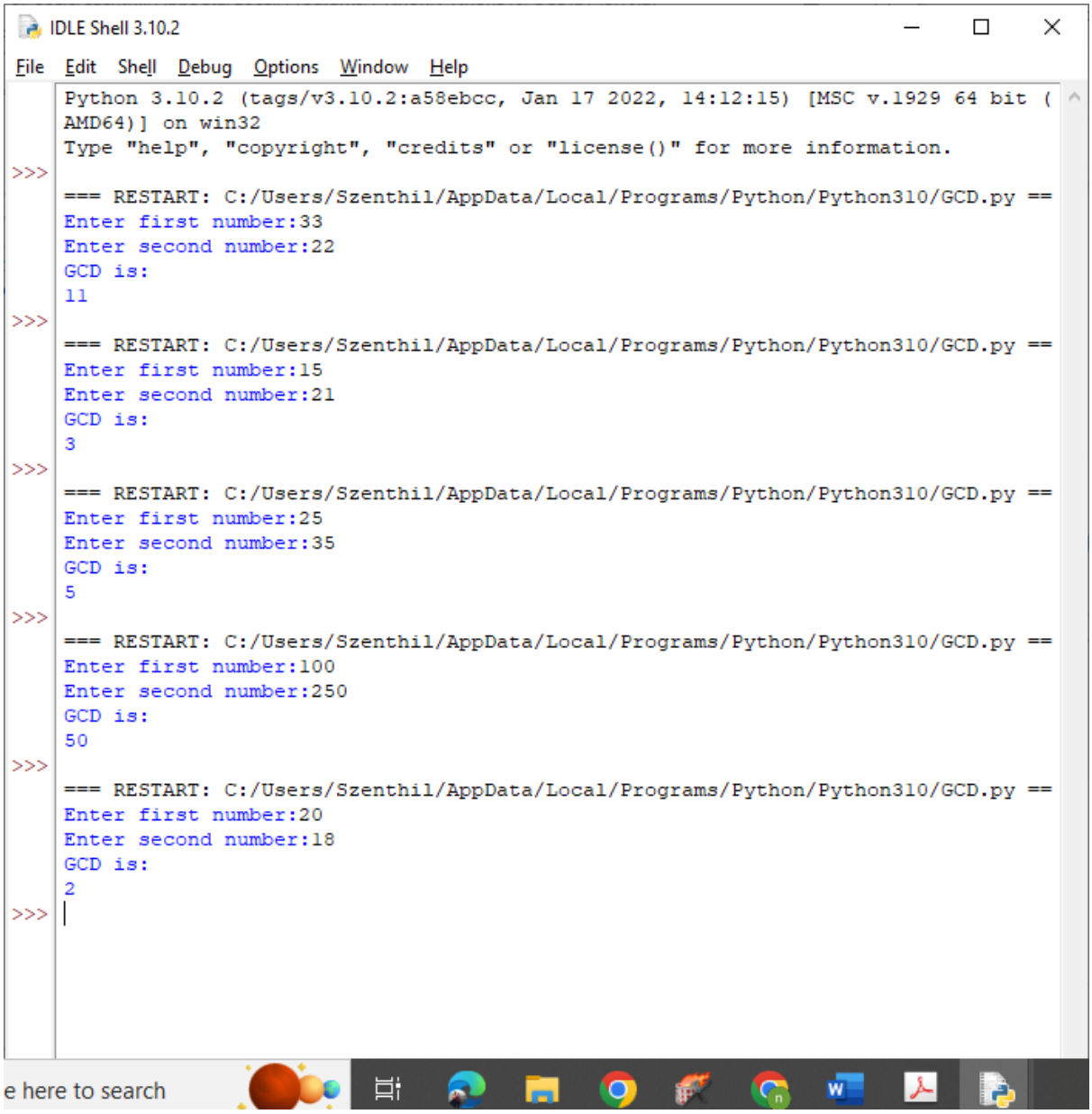
Step9: Print the GCD of given number is b.

Step10: Stop

PROGRAM:

```
def gcd(a,b):
    if(b==0):
        return a
    else:
        return gcd(b,a%b)
a=int(input("Enter first number:"))
b=int(input("Enter second number:"))
GCD=gcd(a,b)
print("GCD is: ")
print(GCD)
```

OUTPUT:



RESULT:

Thus, the python program for GCD is successfully completed.

Program No: 1 F File Name: Ex. No: Date: _____	STRINGS IN PYTHON

AIM:

To write a Python Program to compute the operations and functions of strings in python.

ALGORITHM:

Step1: Start

Step2: Read the value of text.

Step3: lower() – Converts all upper case characters in a string in to lowercase.

Step4: upper() – Converts all lower case characters in a string in to uppercase.

Step5: title() – Convert string to title case.

Step6: swapcase() - Swap the cases of all character in to a string.

Step7: capitalize() – Convert the first character of a string to uppercase.

Step8: Read the function.

Step9: Print the result.

Step10: Stop

PROGRAM:

```
text = “WelcoMe To PytHON”

# string to upper case

print("\nConverted String:")

print(text.upper())

 # string to lower case

print("\nConverted String:")

print(text.lower())
```

```
# title upper case and rest to lower case
print("\nConverted String:")
print(text.title())
# upper case character to lowercase and vice versa
print("\nConverted String:")
print(text.swapcase())
# convert the first character of a string to uppercase
print("\nConverted String:")
print(text.capitalize())
# original string never changes
print("\nOriginal String")
print(text)
```

OUTPUT:

```
IDLE Shell 3.10.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/strings.py

Converted String:
WELCOME TO PYTHON

Converted String:
welcome to python

Converted String:
Welcome To Python

Converted String:
wELCOmE tO pYThon

Converted String:
Welcome to python

Original String
WelcoMe To PytHON
>>>
Ln: 3  Col: 0
```

RESULT:

Thus, the python program to compute the operations and functions of strings in python has been successfully completed.

Program No: 2 A File Name: Ex. No: Date: _____	LISTS IN PYTHON

AIM:

To write a Python Program to compute the methods and functions of python lists.

ALGORITHM:

Step1: Start

Step2: Read the list value of a.

Step3: append() – Adds an element to the end of the list..

Step4: copy() – Returns the shallow copy of the list.

Step5: count() – Returns the number of times a specified element appears in the list.

Step6: extend() – Adds elements from another list to the end of the current list.

Step7: index() – Returns the index of the first occurrence of a specified element.

Step8: insert() – Inserts an element at a specified position.

Step9: pop() – Removes and returns the element at the specified position (or the last element if no index is specified).

Step10: remove() – Removes the first occurrence of a specified element.

Step11: reverse() – Reverses the order of the elements in the list.

Step12: sort() – Sorts the list in ascending order (by default).

Step13: clear() – Removes all elements from the list.

Step14: Read the function.

Step15: Print the result.

Step16: Stop

PROGRAM:

```
a = [1, 2, 3, 4, 5, 6]

# Add 7 to the end of the list

a.append(7)

print(a)

# Create a copy of the list

b = a.copy()

print(b)

# Count occurrences of 5 in the list

print(a.count(5))

# Extend list a by adding elements from list [8, 9]

a.extend([8, 9])

print(a)

# Find the index of 7 in the list

print(a.index(7))

# Insert 2 at index 1

a.insert(1, 2)

print(a)

# Remove and return the last element in the list

a.pop()

print(a)

# Remove the first occurrence of 2
a.remove(2)

print(a)
```

```
# Reverse the list order
a.reverse()
print(a)

# Sort the list in ascending order
a.sort()
print(a)

# Remove all elements from the list
a.clear()
print(a)
```

OUTPUT:

```
IDLE Shell 3.10.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
[1, 2, 3, 4, 5, 6, 7]
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
[1, 2, 3, 4, 5, 6, 7]
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
1
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
[1, 2, 3, 4, 5, 6, 7, 8, 9]
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
6
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
6
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
[1, 2, 2, 3, 4, 5, 6, 7, 8, 9]
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
[1, 2, 3, 4, 5, 6, 7, 8]
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
[1, 2, 3, 4, 5, 6, 7, 8]
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
[8, 7, 6, 5, 4, 3, 2, 1]
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
[1, 2, 3, 4, 5, 6, 7, 8]
>>>
= RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/listsl.py =
[]
>>>
Ln: 39  Col: 0
```

RESULT:

Thus, the python program to compute the methods and functions of python list has been successfully completed.

Program No: 2 B File Name: Ex. No: Date: _____	TUPLES IN PYTHON

AIM:

To write a Python Program to compute the methods and functions of python tuples.

ALGORITHM:

Step1: Start

Step2: Read the tuple value of variable tuple.

Step3: length() – Returns the total length of the tuple.

Step4: Concatenation (+) – It allows for the creation of new tuples by combining the existing ones.

Step5: Repetition(*) – Returns multiple copies of the same tuple.

Step6: maximum() – Returns the largest element from the tuple.

Step7: minimum() – Returns the smallest element from the tuple.

Step8: index() – Accessing the specific element from the tuple by using its position or index number.

Step9: count() – Returns the number of times a specified value appears in the tuple.

Step10: delete() – Deletes the whole tuple.

Step11: Read the function.

Step12: Print the result.

Step13: Stop

PROGRAM:

```
tuple = (10, 20, 30, 40, 50, 60)
# Find the length of the tuple
print(len(tuple))
```

```
# Updating the tuples by using concatenation (+) operator
tuple1 = (10, 20, 30, 40, 50, 60)
tuple2 = (70, 80, 90)
tuple3 = tuple1+tuple2
print(tuple3)

# To make multiple copies of a tuple using repetition (*) operator
print(tuple*2)

# To find the maximum value in the given tuple using max() function
print(max(tuple))

# To find the minimum value in the given tuple using min() function
print(min(tuple))

# To access the specific element from the given tuple by using slice operator [:]
print(tuple.index(50))

# To find the count of specified element the given tuple using count() function
print(tuple.count(10))

# To delete the entire tuple by using del() function
del tuple
print(tuple)
```

OUTPUT:

```
IDLE Shell 3.10.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/tuple.py =
6
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/tuple.py =
(10, 20, 30, 40, 50, 60, 70, 80, 90)
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/tuple.py =
(10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/tuple.py =
60
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/tuple.py =
10
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/tuple.py =
4
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/tuple.py =
1
>>>
== RESTART: C:/Users/Szenthil/AppData/Local/Programs/Python/Python310/tuple.py =
<class 'tuple'>
>>>
Ln: 27  Col: 0
```

RESULT:

Thus, the python program to compute the methods and functions of python tuples has been successfully completed.

Program No: 2 C File Name: Ex. No: Date: _____	DICTIONARIES IN PYTHON

AIM:

To write a Python Program to compute the methods and functions of python dictionary.

ALGORITHM:

Step1: Start

Step2: Read the tuple value of variable a.

Step3: Creating dictionary – It consists of (Key: Value) pairs separates with coma (,) and encloses in curly brackets {}. Each key is separated from it value by a colon (:).

Step4: Accessing dictionary – To access dictionary we can use square brackets along with the key to obtain its value.

Step5: Adding in dictionary – To add dictionary we can use square brackets along with the key to obtain its value.

Step6: Updating dictionary – We can update a dictionary by adding a new entry or a key-value pair, modifying an existing entry, or deleting an existing entry.

Step7: Deletion in dictionary – We can remove individual dictionary elements or clear the entire contents of a dictionary by using clear() function.

Step8: length() – Returns the total length of the dictionary.

Step9: copy() – Allows to copy the key-value pairs in to another variable..

Step10: clear() – Delete the specified key-values and also delete entire dictionary.

Step11: pop() – Deletes an item from the dictionary and it will return the value of a key which is removed.

Step12: popitem() – Removes the recently inserted item. The value of removed key-value pair will be returned.

Step13: Read the function.

Step14: Print the result.

Step15: Stop

PROGRAM:

```
# Creating dictionary

dict = {'Name' : 'Szenthil', 'Roll no' : 123, 'Department' : 'MBA'}

print(dict)

# Accessing dictionary

print(dict['Name'])

print(dict['Roll no'])

print(dict['Department'])

# Adding dictionary

dict['Course'] = 'Business Analytics'

print(dict)

# Updating dictionary

dict['Course'] = 'Business Forecasting'

print(dict)

# Deleting dictionary

del dict['Department']

print(dict)

# Find the length of the dictionary

print(len(dict))

# Copy the key-value pairs in to another variable

dict1=dict.copy()

print(dict1)

# Remove all key-value pairs

dict.clear()
```

```
print(dict)

# Delete any item from the dictionary

dict.pop('Department')

print(dict)

# Delete recently added item from the dictionary

dict.popitem()

print(dict)
```

OUTPUT:

```
IDLE Shell 3.10.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (AMD
64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
{'Name': 'Szenthil', 'Roll no': 123, 'Department': 'MBA'}
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
Szenthil
123
MBA
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
{'Name': 'Szenthil', 'Roll no': 123, 'Department': 'MBA', 'Course': 'Business Analy
tics'}
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
{'Name': 'Szenthil', 'Roll no': 123, 'Department': 'MBA', 'Course': 'Business Forec
asting'}
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
{'Name': 'Szenthil', 'Roll no': 123}
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
3
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
{'Name': 'Szenthil', 'Roll no': 123, 'Department': 'MBA'}
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
{}
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
{'Name': 'Szenthil', 'Roll no': 123}
>>>
== RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\dicti.py =
{'Name': 'Szenthil', 'Roll no': 123}
>>>
Ln: 35  Col: 0
```

RESULT:

Thus, the python program to compute the methods and functions of python dictionary has been successfully completed.

Program No: 3 File Name: Ex. No: Date: _____	ILLUSTRATION ON LAMBDA AND FILTERS IN PYTHON

AIM:

To study and implement the use of **lambda functions** and **filter functions** in Python for concise and efficient data processing.

ALGORITHM:

Step 1: Start the program.

Step 2: Define a list of elements (e.g., numbers).

Step 3: Create a lambda function to define the desired operation or condition.

Step 4: Apply the lambda function:

- Directly for simple calculations, or
- Inside the `filter()` function to select specific elements.

Step 5: Store the result in a variable.

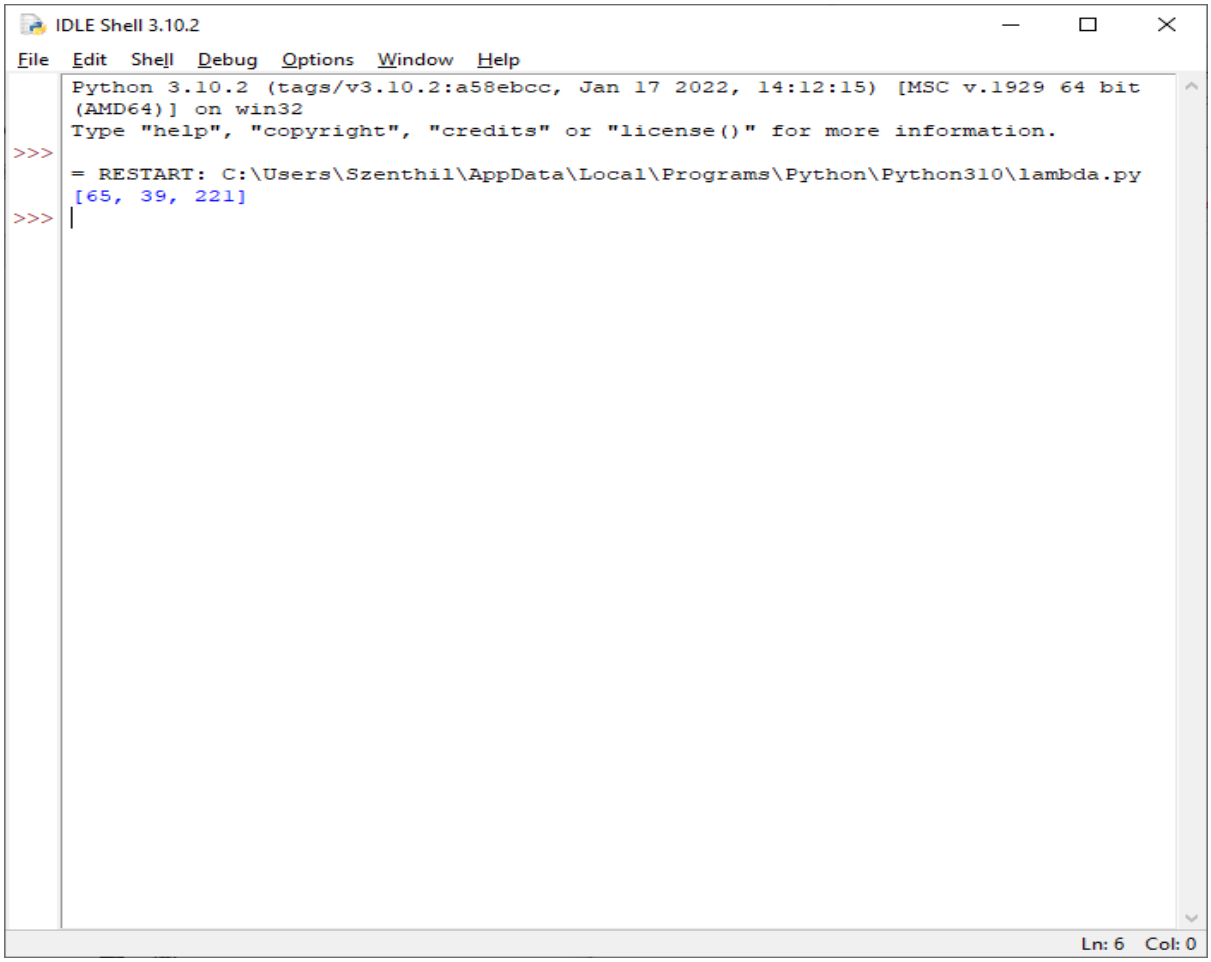
Step 6: Display the output.

Step 7: Stop the program.

PROGRAM:

```
a = [12, 65, 54, 39, 102, 339, 221, 50, 70]

# Using filter() with lambda to get numbers divisible by 13

res = list(filter(lambda x: (x % 13 == 0), a))


# Printing the result
print(res)
```

OUTPUT



RESULT:

Thus, the python program for implementing **lambda functions** and **filter functions** has been successfully completed.

Program No: 4 File Name: Ex. No: Date: _____	IMPLEMENTING INHERRITANCE

AIM:

To study and implement **single inheritance in Python**, where a child class inherits properties and methods from a parent class.

ALGORITHM:

Step 1: Start the program.

Step 2: Define a parent class Animal.

Step 3: Create a constructor __init__ in Animal to initialize the attribute name.

Step 4: Define a method info() in the parent class to display the animal's name.

Step 5: Define a child class Dog that inherits from Animal.

Step 6: Create a constructor __init__ in Dog with parameters name and breed.

Step 7: Use super().__init__(name) to call the parent class constructor.

Step 8: Initialize the additional attribute breed in the child class.

Step 9: Define a method details() in Dog to display the dog's name and breed.

Step 10: Create an object of the Dog class.

Step 11: Call the parent class method info() using the object.

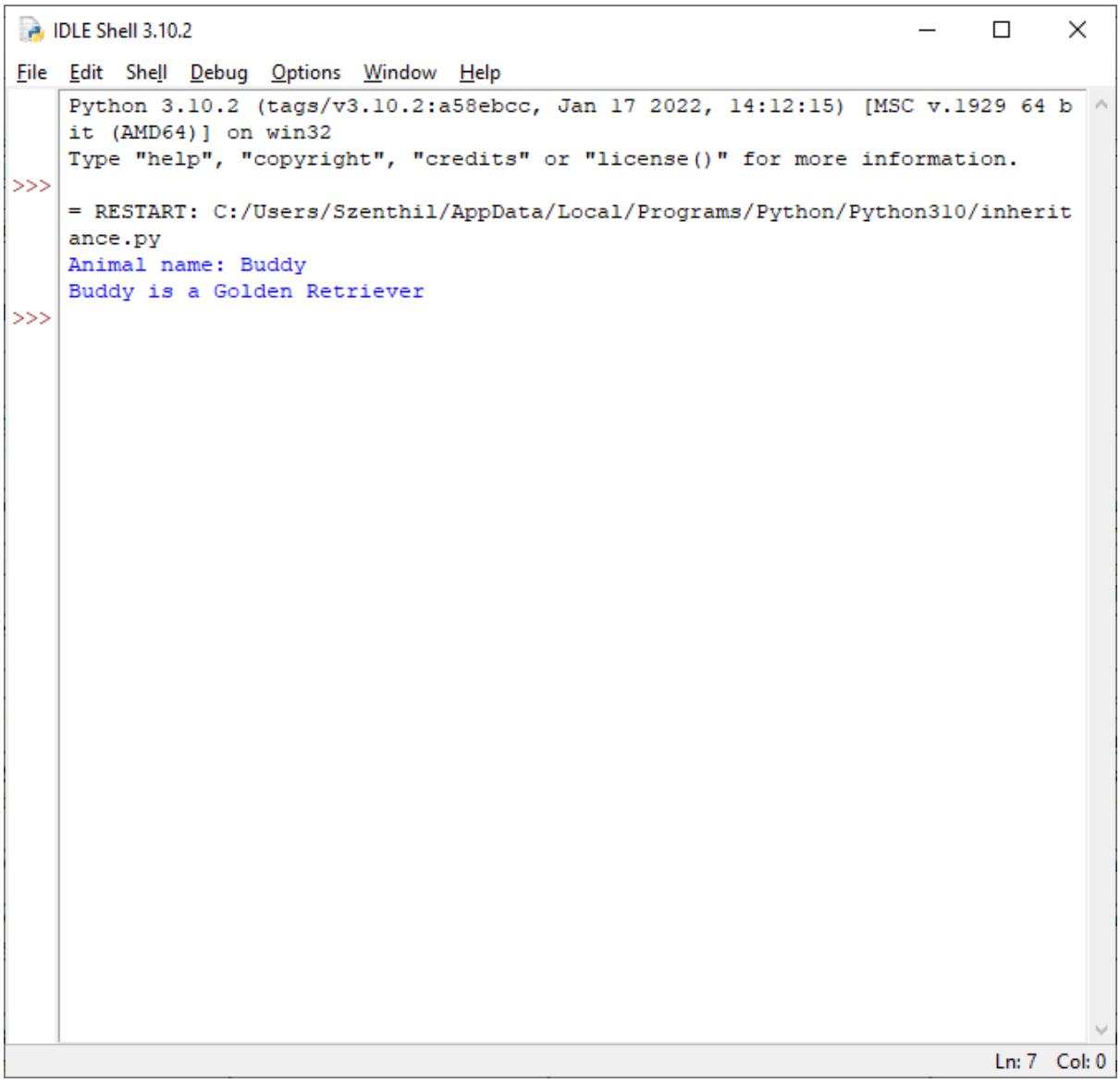
Step 12: Call the child class method details() using the object.

Step 13: Stop the program.

PROGRAM:

```
# Parent Class: Animal
class Animal:
    def __init__(self, name):
        self.name = name

    def info(self):
        print("Animal name:", self.name)

# Child Class: Dog
class Dog(Animal):
    def __init__(self, name, breed):
        super().__init__(name)   # Call parent constructor
        self.breed = breed

    def details(self):
        print(self.name, "is a", self.breed)

d = Dog("Buddy", "Golden Retriever")
d.info()      # Parent method
d.details()   # Child method
```

OUTPUT



RESULT:

Thus, the python program to compute **single inheritance in Python**, where a child class inherits properties and methods from a parent class has been successfully completed.

Program No: 5 File Name: Ex. No: Date: _____	IMPLEMENTING METHOD OVERLOADING

AIM:

To study and implement **method overloading in Python** using variable-length arguments (*args), since Python does not support traditional method overloading.

ALGORITHM:

Step 1: Start the program.

Step 2: Define a function `add(*args)` to accept a variable number of arguments.

Step 3: Initialize a variable `res` to 0.

Step 4: Iterate through each argument using a loop.

Step 5: Add each argument to `res`.

Step 6: Return or print the result.

Step 7: Call the function with different numbers of arguments:

- Two arguments
- Three arguments
- More arguments

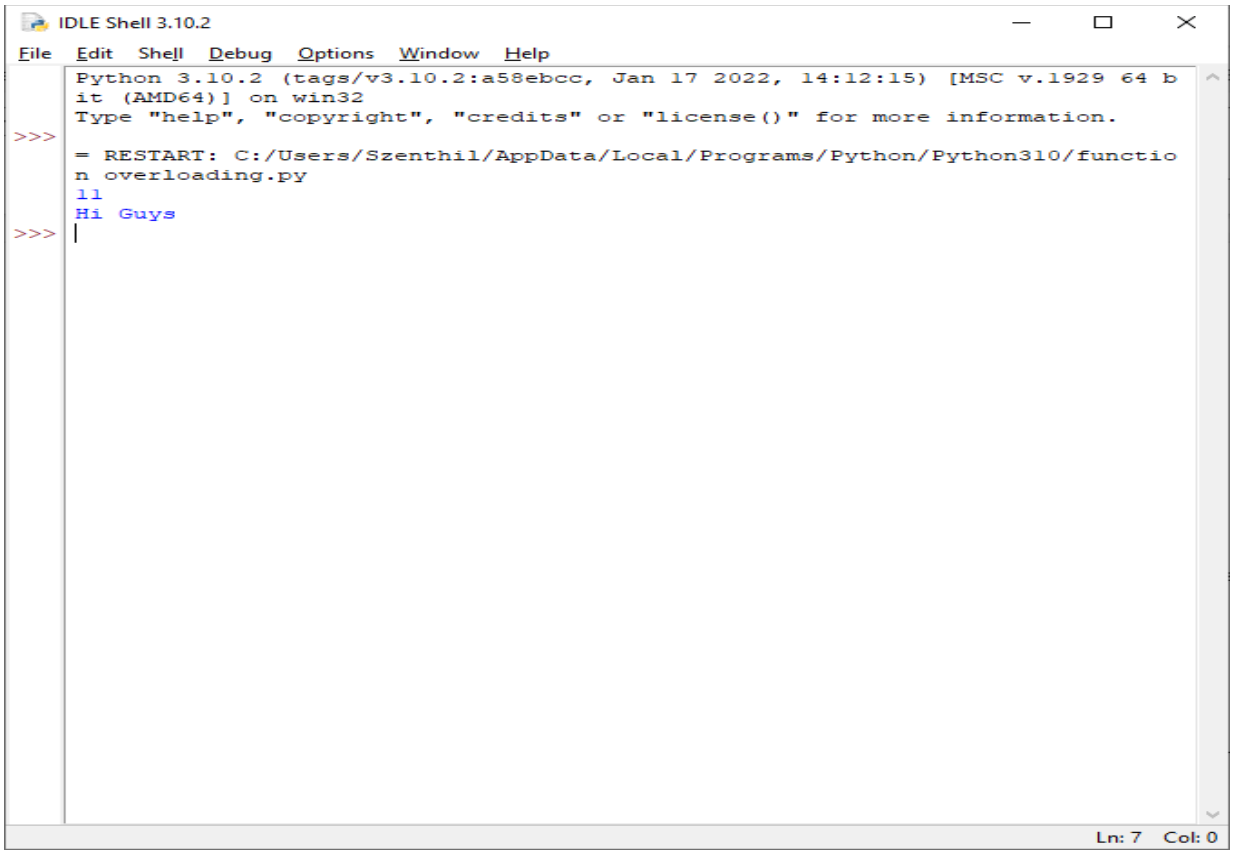
Step 8: Display the results.

Step 9: Stop the program.

PROGRAM:

```
def add(datatype, *args):
    if datatype == 'int':
        res = 0
    elif datatype == 'str':
        res = ''

    for item in args:
        res += item

    print(res)

add('int', 5, 6)
add('str', 'Hi ', 'Guys')
```

OUTPUT



RESULT:

Thus, the python program to import CSV data files using pandas has been successfully completed.

Program No: 6 File Name: Ex. No: Date: _____	ILLUSTRATION ON HOW TO RAISE AN EXCEPTION

AIM:

To study and implement **exception handling in Python** to handle runtime errors and ensure smooth program execution.

ALGORITHM:

Step 1: Start the program.

Step 2: Write the code that may produce an error inside a try block.

Step 3: Add one or more except blocks to handle specific exceptions.

Step 4: Optionally, include an else block to execute code when no exception occurs.

Step 5: Include a finally block to execute code that must run regardless of exceptions.

Step 6: Execute the program.

Step 7: Display appropriate output based on whether an exception occurred or not.

Step 8: Stop the program.

Step 9 :

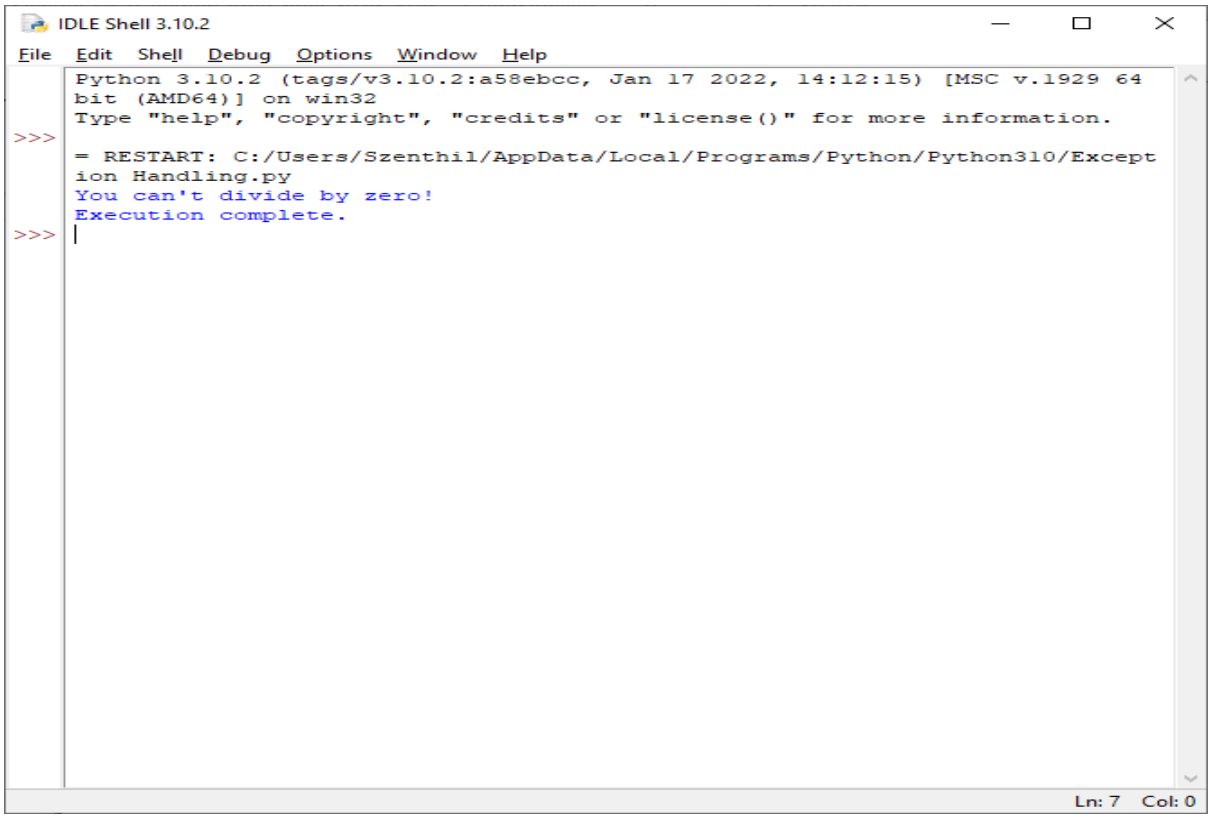
Explanation

- try: Contains risky code
- except: Handles errors
- else: Runs if no error occurs
- finally: Always executes

PROGRAM:

```
try:
    n = 0
    res = 100 / n

except ZeroDivisionError:
    print("You can't divide by zero!")

except ValueError:
    print("Enter a valid number!")

else:
    print("Result is", res)

finally:
    print("Execution complete.")
```

OUTPUT



RESULT:

Thus, the python program to compute the **exception handling in Python** to handle runtime errors has been successfully completed.

Program No: 7 File Name: Ex. No: Date: _____	IMPLEMENTING MODULES [CREATING AND IMPORTING]

AIM:

To create a Python module with user-defined functions and import it into another program to perform operations like addition, subtraction, and displaying greetings.

ALGORITHM:

Step 1: Start the program.

Step 2: Create a module file named `mymodule.py`.

Step 3: Define functions inside the module:

- `add(a, b)` to perform addition
- `sub(a, b)` to perform subtraction
- `greet(name)` to return a greeting message

Step 4: Save the module file.

Step 5: Create the main program file `main.py`.

Step 6: Import the module using `import mymodule`.

Step 7: Call the functions from the module using:

- `mymodule.add()`
- `mymodule.sub()`
- `mymodule.greet()`

Step 8: Display the results.

Step 9: Stop the program.

PROGRAM:

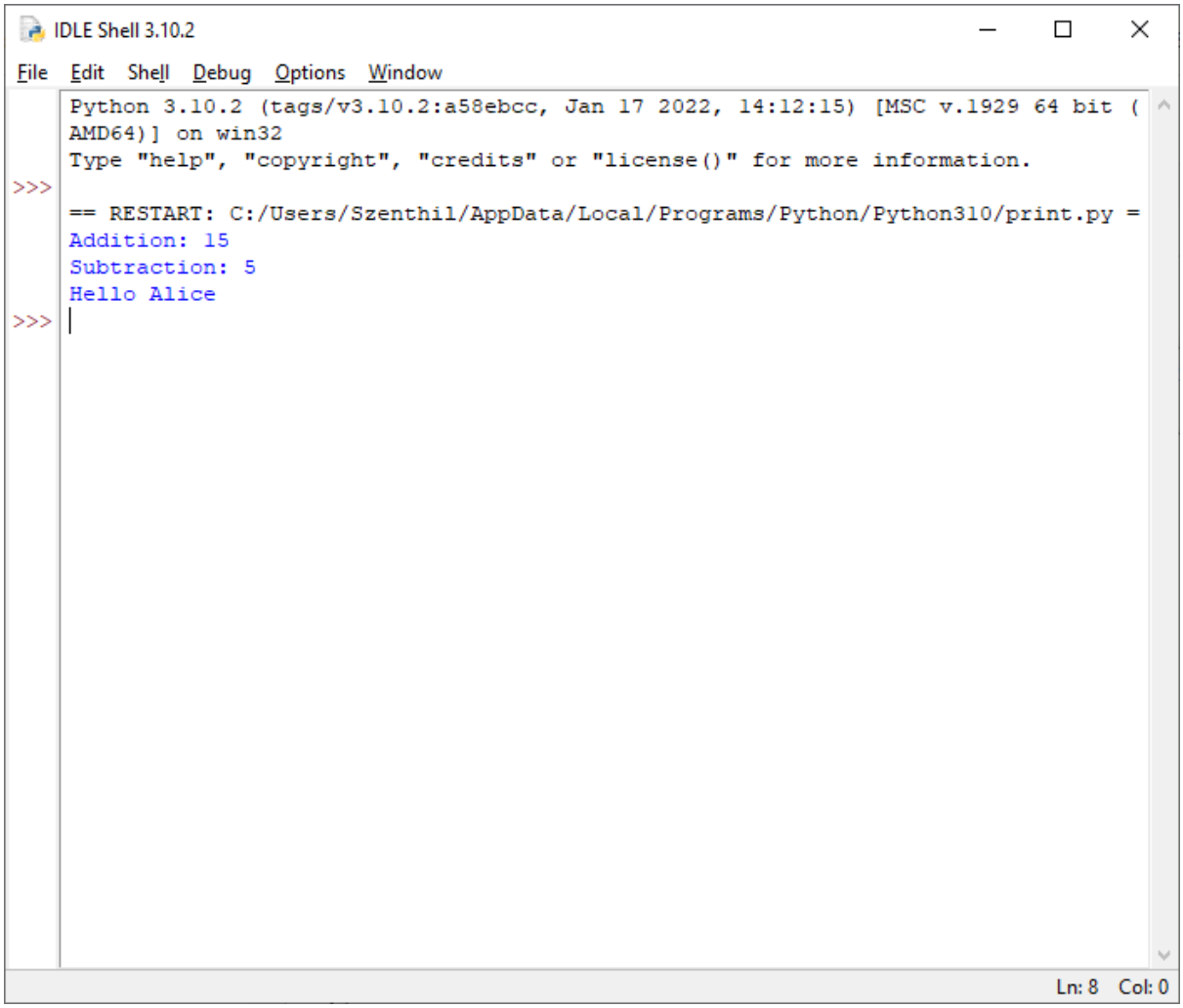
Create Module File (mymodule.py)

```
# mymodule.py

def add(a, b):
    return a + b

def sub(a, b):
    return a - b

def greet(name):
    return "Hello " + name
```

Main Program (main.py)

```
# main.py

import mymodule

# Using functions from module
print("Addition:", mymodule.add(10, 5))
print("Subtraction:", mymodule.sub(10, 5))
print(mymodule.greet("Alice"))
```

OUTPUT:



RESULT:

Thus, the python program to compute Python module with user-defined functions and import it into another program has been successfully completed.

Program No: 8 File Name: Ex. No: Date: _____	IMPLEMENTING THREADS

AIM:

To study and implement **multithreading in Python with arguments**, where a thread executes a function with parameters and runs concurrently with the main program.

ALGORITHM:

Step 1: Start the program.

Step 2: Import the required modules: `threading` and `time`.

Step 3: Define a function `my_function(argument1, argument2)` that:

- Prints the received arguments
- Simulates a task using `time.sleep()`
- Displays completion message

Step 4: Create a thread object using `threading.Thread()`.

Step 5: Pass the function as the `target` and arguments using the `args` parameter.

Step 6: Start the thread using the `start()` method.

Step 7: Execute statements in the main thread (parallel execution).

Step 8: Use `join()` to wait for the thread to finish execution.

Step 9: Display the final completion message.

Step 10: Stop the program.

PROGRAM:

```
import threading
import time

def my_function(argument1, argument2):
    print(f"Thread started with args: {argument1} and {argument2}")
    time.sleep(2) # Simulate some I/O work
    print("Thread finished its task.")

# Create and start the thread
my_thread = threading.Thread(target=my_function, args=('Hello', 'World',))
my_thread.start()

print("Main thread: Doing other work.")

# Wait for the thread to finish
my_thread.join()
print("Main thread: All threads are complete. Exiting program.")
```

OUTPUT

```
IDLE Shell 3.10.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.10.2 (tags/v3.10.2:a58ebcc, Jan 17 2022, 14:12:15) [MSC v.1929 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Szenthil\AppData\Local\Programs\Python\Python310\threads.py
Main thread: Doing other work.Thread started with args: Hello and World

Thread finished its task.
Main thread: All threads are complete. Exiting program.
>>>
Ln: 9  Col: 0
```

RESULT:

Thus, the program to compute **multithreading in Python with arguments**, where a thread executes a function with parameters has been successfully completed.

Program No: 9 A File Name: Ex. No: Date: _____	ILLUSTRATION ON COMMAND LINE ARGUMENTS USING PYTHON

AIM:

To study and implement **command line arguments in Python** to calculate the sum of numbers passed during program execution.

ALGORITHM:

Step 1: Start the program.

Step 2: Import the sys module.

Step 3: Access the list of command line arguments using sys.argv.

Step 4: Display:

- Total number of arguments using len(sys.argv)
- Script name using sys.argv[0]
- Input arguments using sys.argv[1:]

Step 5: Initialize a variable total = 0.

Step 6: Use a loop to iterate through each argument in sys.argv[1:].

Step 7: Convert each argument to integer using int().

Step 8: Add each value to total.

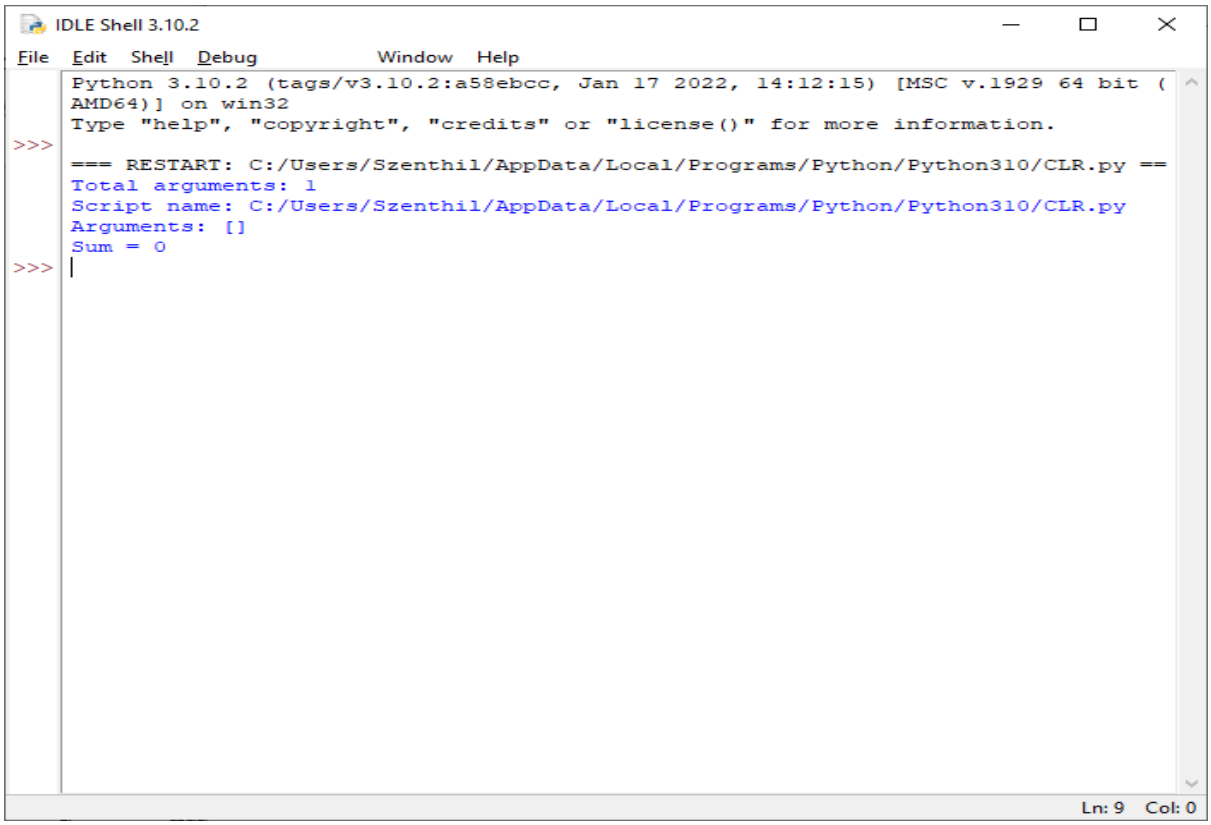
Step 9: Display the sum.

Step 10: Stop the program.

PROGRAM:

```
import sys

print("Total arguments:", len(sys.argv))
print("Script name:", sys.argv[0])
print("Arguments:", sys.argv[1:])

total = 0

for arg in sys.argv[1:]:

    total += int(arg)

print("Sum =", total)
```

OUTPUT:



RESULT:

Thus, the program to compute **command line arguments in Python** to calculate the sum of numbers passed during program execution has been successfully completed.

Program No: 9 B File Name: Ex. No: Date: _____	ILLUSTRATION ON REGULAR EXPRESSIONS IN PYTHON

AIM:

To study and implement **regular expressions in Python** for pattern matching and text processing.

ALGORITHM:

Step 1: Start the program.

Step 2: Import the `re` module.

Step 3: Define a string on which operations will be performed.

Step 4: Specify a pattern (word or sequence to search).

Step 5: Use a RegEx function such as:

- `re.search()` → finds first match
- `re.match()` → checks match at beginning
- `re.findall()` → returns all matches

Step 6: Store the result in a variable.

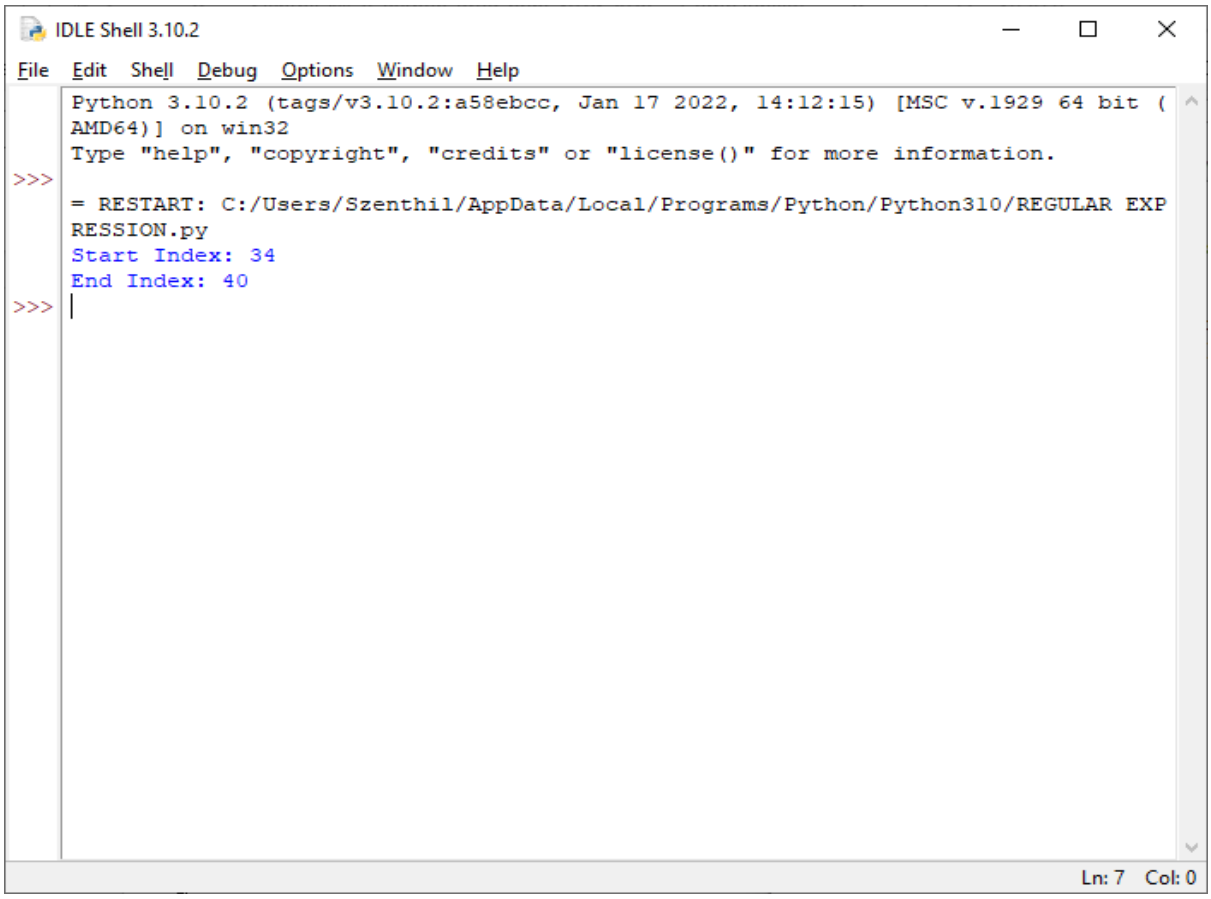
Step 7: Display the result.

Step 8: Stop the program.

PROGRAM:

```
import re

s = 'GeeksforGeeks: A computer science portal for geeks'
match = re.search(r'portal', s)

print('Start Index:', match.start())
print('End Index:', match.end())
```

OUTPUT:



RESULT:

Thus, the program for **regular expressions in Python** was successfully executed.

Program No: 10 File Name: Ex. No: Date: _____	DESCRIPTIVE STATISTICS USING NUMPY

AIM:

To study and implement **statistical measures in Python** using NumPy and SciPy, including measures of central tendency and dispersion.

ALGORITHM:

Step 1: Start the program.

Step 2: Import required libraries:

- numpy for numerical operations
- scipy.stats for mode calculation

Step 3: Define a dataset using a NumPy array.

Step 4: Calculate **measures of central tendency**:

- Mean using np.mean()
- Median using np.median()
- Mode using stats.mode()

Step 5: Calculate **measures of dispersion**:

- Minimum using np.min()
- Maximum using np.max()
- Range using np.ptp()
- Variance using np.var()
- Standard deviation using np.std()

Step 6: Calculate **quartiles**:

- First quartile (Q1) using np.percentile(data, 25)
- Third quartile (Q3) using np.percentile(data, 75)

Step 7: Compute **Interquartile Range (IQR)**:

- IQR = Q3 - Q1

Step 8: Display all calculated values.

Step 9: Stop the program.

PROGRAM:

```
import numpy as np
from scipy import stats # Required for the mode calculation

# Sample data
data = np.array([1, 2, 2, 3, 4, 5, 5, 5, 6, 7])

# Measures of Central Tendency
mean_value = np.mean(data)
median_value = np.median(data)
# Mode requires SciPy
mode_result = stats.mode(data)
mode_value = mode_result.mode[0]

# Measures of Dispersion
min_value = np.min(data)
max_value = np.max(data)
range_value = np.ptp(data) # Peak-to-peak range
variance_value = np.var(data)
std_deviation_value = np.std(data)
q1 = np.percentile(data, 25)
q3 = np.percentile(data, 75)
iqr = q3 - q1

print(f"Mean: {mean_value}")
print(f"Median: {median_value}")
print(f"Mode: {mode_value}")
print(f"Minimum: {min_value}")
```

```
print(f"Maximum: {max_value}")
print(f"Range: {range_value}")
print(f"Variance: {variance_value}")
print(f"Standard Deviation: {std_deviation_value}")
print(f"Interquartile Range (IQR): {iqr}")
```

OUTPUT:

data = [1, 2, 2, 3, 4, 5, 5, 5, 6, 7]

Mean: 4.0

Median: 4.5

Mode: 5

Minimum: 1

Maximum: 7

Range: 6

Variance: 3.4

Standard Deviation: 1.8439088914585775

Interquartile Range (IQR): 3.0

RESULT:

Thus, the program for calculating **statistical measures (mean, median, mode, variance, standard deviation, and IQR)** using Python was successfully executed.

Program No: 11 File Name: Ex. No: Date: _____	ILLUSTRATE INDEXING OPERATIONS IN DATA FRAMES

AIM:

To study and illustrate **indexing operations in Pandas DataFrame** for accessing and manipulating data.

ALGORITHM:

Step 1: Start the program.

Step 2: Import the pandas library.

Step 3: Create a DataFrame with sample data.

Step 4: Perform indexing operations:

- Select a column using df['column_name']
- Select multiple columns using df[['col1', 'col2']]
- Select rows using .loc[] (label-based indexing)
- Select rows using .iloc[] (position-based indexing)

Step 5: Display the selected data.

Step 6: Stop the program.

PROGRAM:

```
import pandas as pd

data = {'Name': ['John', 'Anna', 'Peter', 'Linda'],
        'Age': [28, 24, 35, 32],
        'City': ['New York', 'Paris', 'Berlin', 'London']}
df = pd.DataFrame(data, index=['row1', 'row2', 'row3', 'row4'])
print(df)
```

OUTPUT

	Name	Age	City
row1	John	28	New York
row2	Anna	24	Paris
row3	Peter	35	Berlin
row4	Linda	32	London

RESULT:

Thus, indexing operations in a **Pandas DataFrame** were successfully illustrated.

Program No: 12 File Name: Ex. No: Date: _____	ILLUSTRATE VARIOUS PLOTS USING PANDAS AND MATPLOTLIB

AIM:

To study and illustrate **various plots using Pandas and Matplotlib** for data visualization.

ALGORITHM:

Step 1:

Start the program.

Step 2:

Import required libraries:

- `pandas`
- `matplotlib.pyplot`

Step 3:

Create a dataset using a dictionary.

Step 4:

Convert the dataset into a Pandas DataFrame.

Step 5:

Generate different types of plots:

- Line plot
- Bar plot
- Histogram
- Scatter plot

Step 6:

Display each plot using `plt.show()`.

Step 7:

Stop the program.

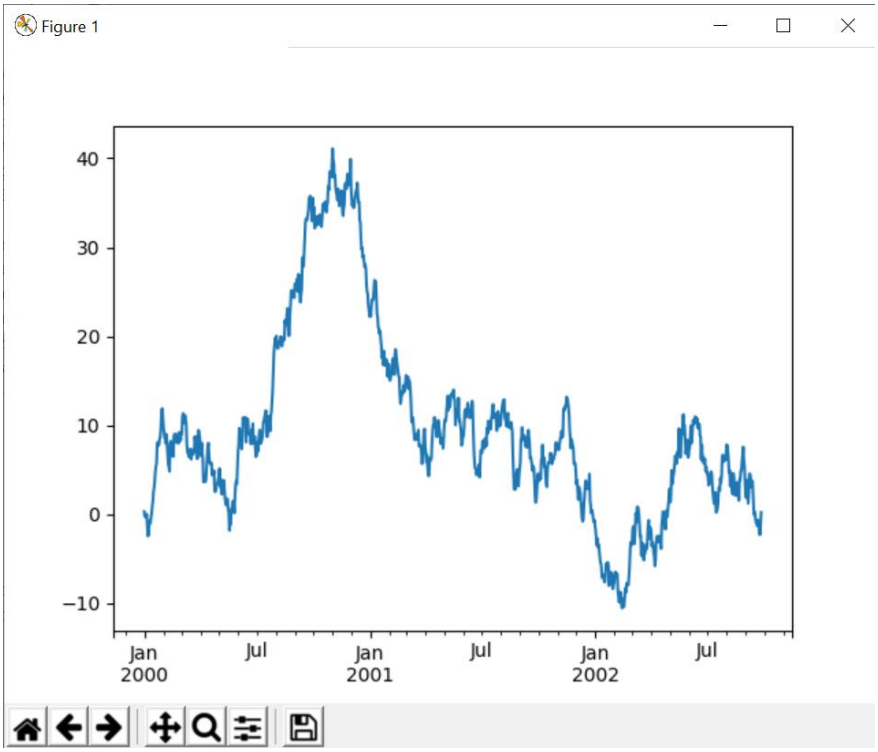
PROGRAM:

pip install matplotlib
pip install pandas
pip install numpy

Basic plotting

```
# import libraries

import matplotlib.pyplot as plt

import pandas as pd

import numpy as np

ts = pd.Series(np.random.randn(1000), index = pd.date_range('1/1/2000', periods = 1000))

ts = ts.cumsum()

ts.plot()

plt.show()
```

OUTPUT

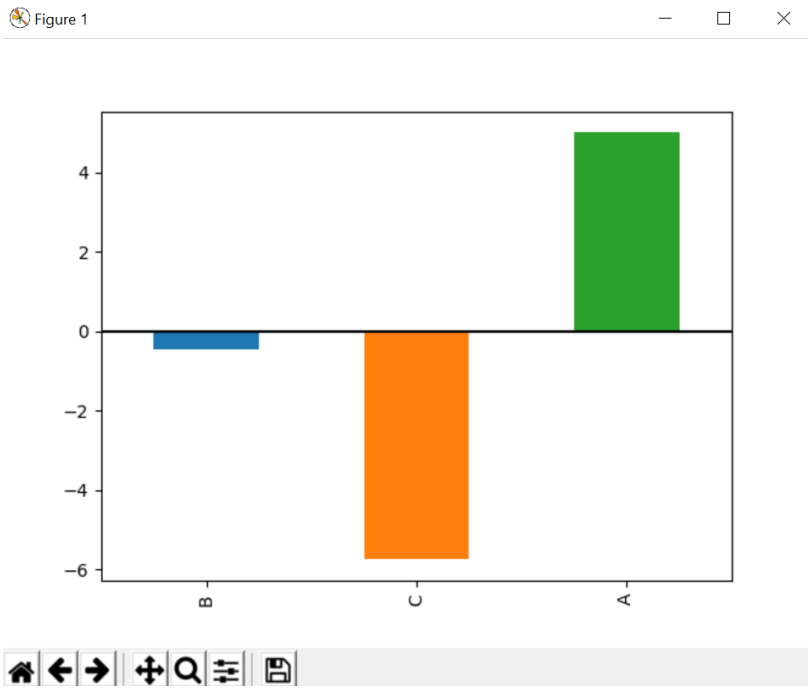


BAR PLOT

PROGRAM:

```
# importing libraries
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
ts = pd.Series(np.random.randn(1000), index = pd.date_range(
                        '1/1/2000', periods = 1000))
df = pd.DataFrame(np.random.randn(1000, 4), index = ts.index,
                        columns = list('ABCD'))
df3 = pd.DataFrame(np.random.randn(1000, 2),
            columns =['B', 'C']).cumsum()
df3['A'] = pd.Series(list(range(len(df))))
df3.iloc[5].plot.bar()
plt.axhline(0, color ='k')
plt.show()
```

OUTPUT

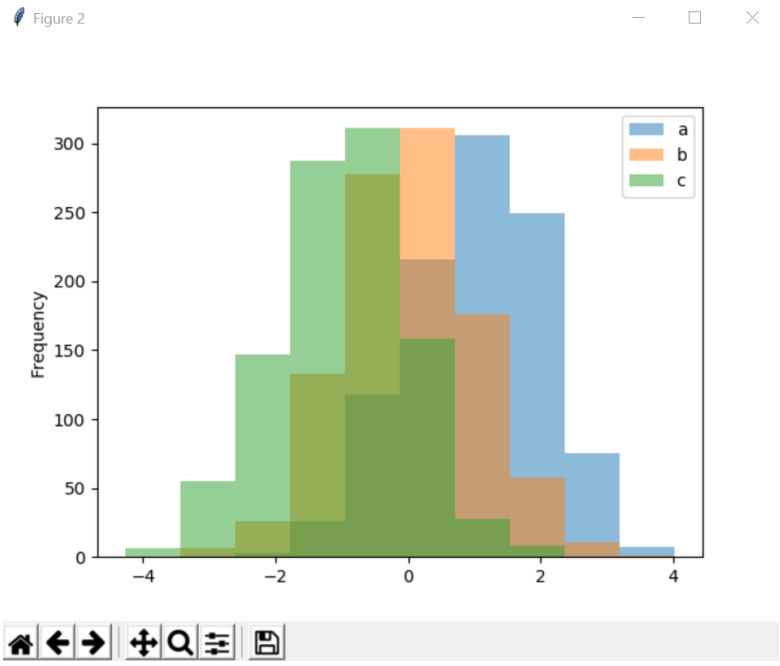


HISTOGRAMS

PROGRAM:

```
# importing libraries

import matplotlib.pyplot as plt

import pandas as pd

import numpy as np


df4 = pd.DataFrame({'a': np.random.randn(1000) + 1,

                    'b': np.random.randn(1000),

                    'c': np.random.randn(1000) - 1},

                         columns =['a', 'b', 'c'])

plt.figure()


df4.plot.hist(alpha = 0.5)

plt.show()
```

OUTPUT

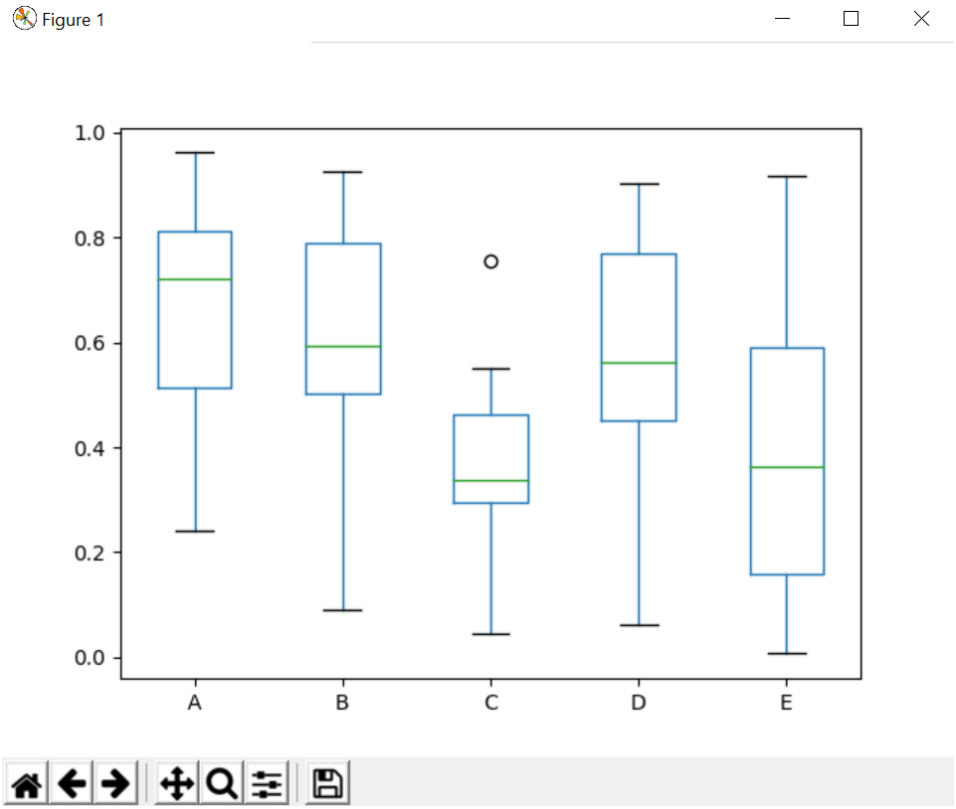


Box plot using Series and matplotlib

PROGRAM:

```
# importing libraries

import matplotlib.pyplot as plt

import pandas as pd

import numpy as np


df = pd.DataFrame(np.random.rand(10, 5),

      columns =['A', 'B', 'C', 'D', 'E'])


df.plot.box()

plt.show()
```

OUTPUT

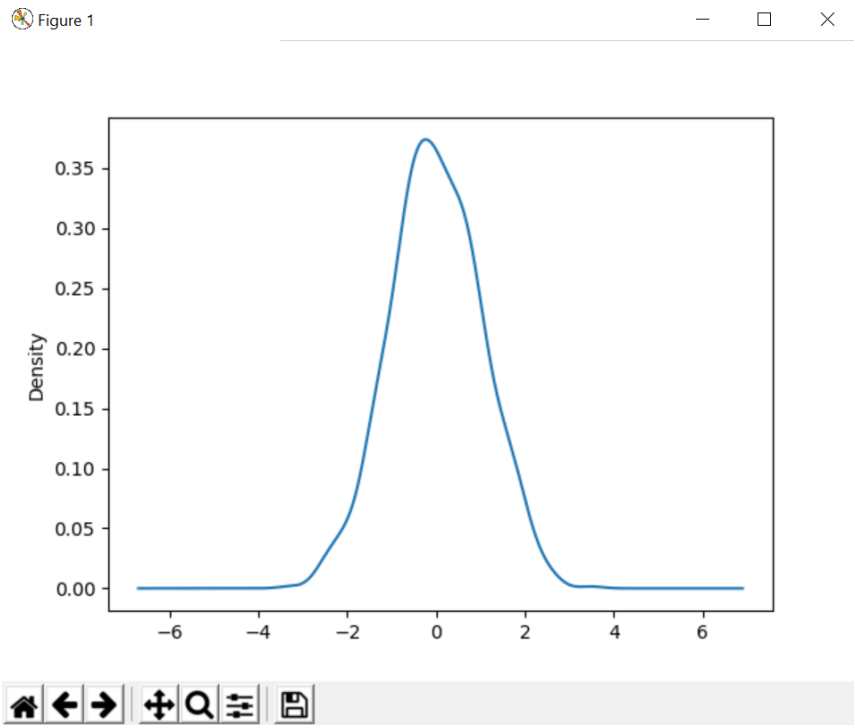


Density plot:

PROGRAM:

```
# importing libraries

import matplotlib.pyplot as plt

import pandas as pd

import numpy as np


df = pd.DataFrame(np.random.rand(10, 5),

      columns =['A', 'B', 'C', 'D', 'E'])


ser = pd.Series(np.random.randn(1000))

ser.plot.kde()

plt.show()
```

OUTPUT

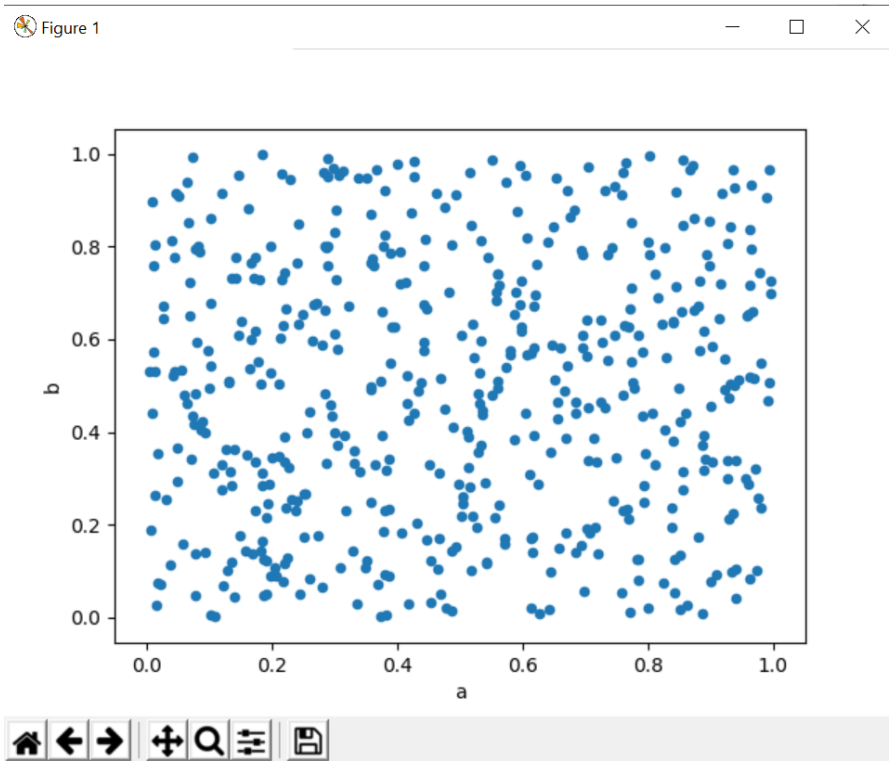


Scatter plot:

PROGRAM:

```
# importing libraries

import matplotlib.pyplot as plt

import pandas as pd

import numpy as np


df = pd.DataFrame(np.random.rand(500, 4),

          columns =['a', 'b', 'c', 'd'])


df.plot.scatter(x ='a', y ='b')

plt.show()
```

OUTPUT

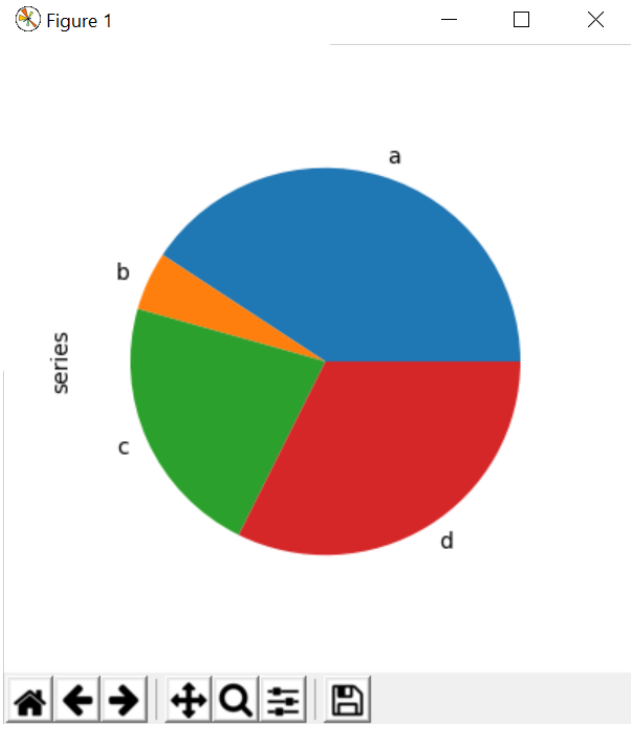


Pie plot:

PROGRAM:

```
# importing libraries
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

series = pd.Series(3 * np.random.rand(4),
  index =['a', 'b', 'c', 'd'], name ='series')

series.plot.pie(figsize =(4, 4))
plt.show()
```

OUTPUT



RESULT:

Thus, various plots using **Pandas and Matplotlib** were successfully illustrated.

Program No: 13 File Name: Ex. No: Date: _____	BUILDING GUI APPLICATIONS WITH TKINTER

AIM:

To study and implement a **basic GUI window using Tkinter** in Python.

ALGORITHM:

Step 1: Start the program.

Step 2:

Import the `tkinter` module.

Step 3:

Create the main window using `Tk()`.

Step 4:

Set the title of the window using `title()`.

Step 5:

Set the window size using `geometry()`.

Step 6:

Keep the window running using `mainloop()`.

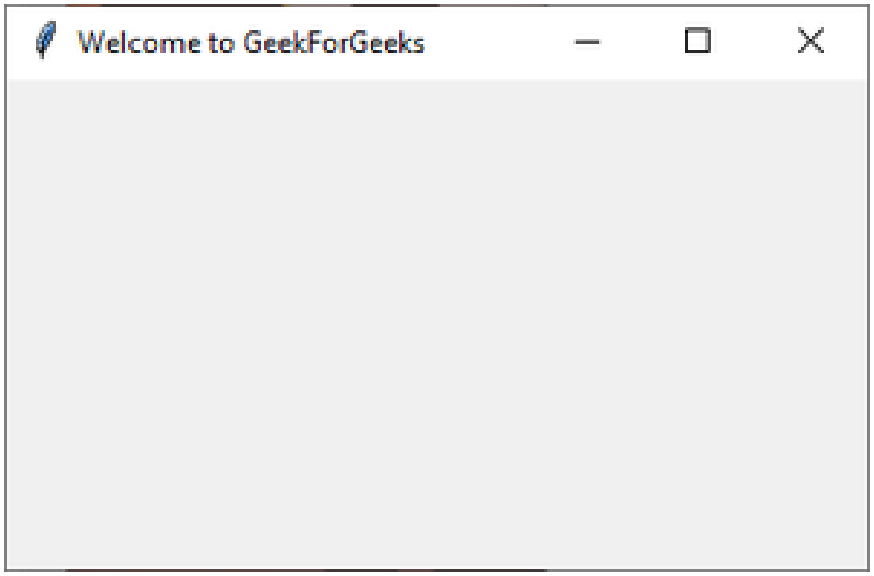
Step 7:

Stop the program when the window is closed.

PROGRAM:

```
# Import Module
from tkinter import *

# create root window
root = Tk()
# root window title and dimension
root.title("Welcome to GeekForGeeks")
# Set geometry (widthxheight)
root.geometry('350x200')
# all widgets will be here
# Execute Tkinter
root.mainloop()
```

OUTPUT



RESULT:

Thus, a **basic Tkinter GUI window** was successfully created and executed.

Program No: 14 File Name: Ex. No: Date: _____	CREATING TABLES USING SQLITE

AIM:

To study and implement **creating tables in SQLite using Python**.

ALGORITHM:

Step 1:

Start the program.

Step 2:

Import the `sqlite3` module.

Step 3:

Create or connect to a database using `connect()`.

Step 4:

Create a cursor object using `cursor()`.

Step 5:

Write an SQL query to create a table.

Step 6:

Execute the query using `execute()`.

Step 7:

Commit the changes using `commit()`.

Step 8:

Close the connection using `close()`.

Step 9:

Stop the program.

PROGRAM:

```
import sqlite3

# Step 1: Connect to database

conn = sqlite3.connect("school.db")

# Step 2: Create cursor

cur = conn.cursor()

# Step 3: Create table

cur.execute("""

CREATE TABLE IF NOT EXISTS student (

    id INTEGER PRIMARY KEY,

    name TEXT NOT NULL,

    marks INTEGER

)

""")

# Step 4: Insert records (ignore if already exists)

students = [

    (1, "John", 85),

    (2, "Alice", 90),
```

```
    (3, "Bob", 78)

]

cur.executemany("INSERT OR IGNORE INTO student VALUES (?, ?, ?)", students)

# Step 5: Commit changes

conn.commit()

# Step 6: Fetch and display data

print("Student Table Data:")

cur.execute("SELECT * FROM student")

for record in cur.fetchall():

    print("ID:", record[0], "| Name:", record[1], "| Marks:", record[2])

# Step 7: Close connection

conn.close()

print("Table created and data displayed successfully!")
```

OUTPUT

```
==================== RESTART: C:/Users/stu
Student Table Data:
ID: 1 | Name: John | Marks: 85
ID: 2 | Name: Alice | Marks: 90
ID: 3 | Name: Bob | Marks: 78
Table created and data displayed successfully!
```

RESULT:

Thus, the program for **creating a table using SQLite in Python** was successfully executed.

Program No: 15 File Name: Ex. No: Date: _____	ILLUSTRATION ON DATABASE CONNECTIVITY

AIM:

To demonstrate database connectivity using SQLite in Python and perform basic operations like creating a table and retrieving data.

ALGORITHM:

Step 1: Start the program.

Step 2: Import the `sqlite3` module.

Step 3: Establish a connection to the database using `connect()`.

Step 4: Create a cursor object using `cursor()`.

Step 5:

Execute SQL queries using `execute()`:

- Create table
- Insert data
- Retrieve data

Step 6:

Fetch data using `fetchall()` or `fetchone()`.

Step 7:

Display the results.

Step 8:

Commit changes using `commit()`.

Step 9:

Close the connection using `close()`.

Step 10:

Stop the program.

PROGRAM:

```
import sqlite3

# Step 3: Connect to database

conn = sqlite3.connect("demo.db")

# Step 4: Create cursor

cursor = conn.cursor()

# Step 5: Create table and insert data

cursor.execute("""

CREATE TABLE IF NOT EXISTS employee (

    emp_id INTEGER PRIMARY KEY,

    emp_name TEXT NOT NULL,

    department TEXT,

    salary REAL

)

""")

employees = [

    (1, "John Doe", "HR", 50000),

    (2, "Alice Smith", "IT", 60000),
```

```
    (3, "Bob Johnson", "Sales", 55000)

]

cursor.executemany("INSERT OR IGNORE INTO employee VALUES (?, ?, ?, ?)",
employees)

# Step 6: Retrieve and display data

cursor.execute("SELECT * FROM employee")

rows = cursor.fetchall()

print("Employee Table Data:")

for row in rows:

    print(f"ID: {row[0]} | Name: {row[1]} | Dept: {row[2]} | Salary: {row[3]}")

# Step 7: Commit and close connection

conn.commit()

conn.close()
```

OUTPUT

```
>>>
========== RESTART: C:/Users/student/Desktop/ex15.py ========
Employee Table Data:
ID: 1 | Name: John Doe | Dept: HR | Salary: 50000.0
ID: 2 | Name: Alice Smith | Dept: IT | Salary: 60000.0
ID: 3 | Name: Bob Johnson | Dept: Sales | Salary: 55000.0
>>>
```

RESULT:

Thus, the program for **database connectivity using SQLite in Python** was successfully implemented and executed.